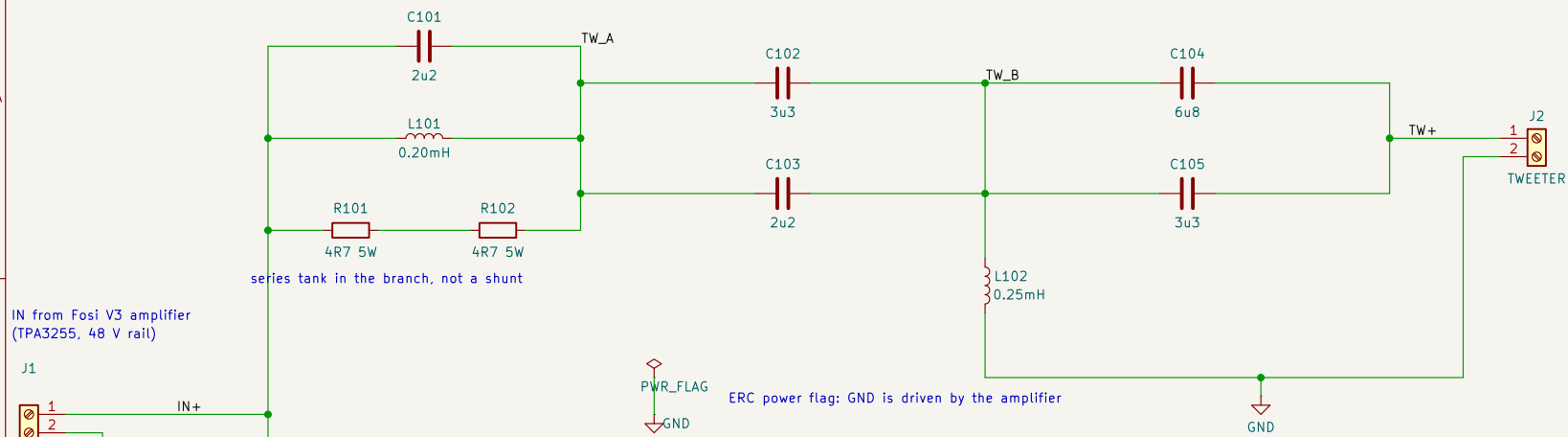
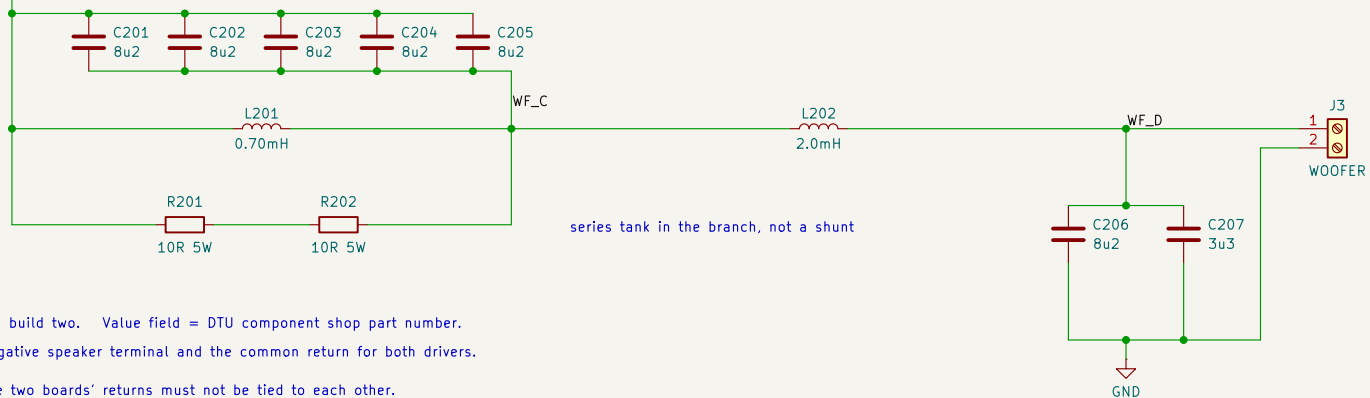


TWEETER BRANCH – Dayton RST28F-4 (4 ohm)



WOOFER BRANCH – Dayton RS180P-8 (8 ohm)



NOTES – one board per speaker, build two. Value field = DTU component shop part number.

1. GND = IN- , the amplifier's negative speaker terminal and the common return for both drivers.

It is NOT chassis earth, and the two boards' returns must not be tied to each other.

2. All capacitors are FILM and must be rated ≥ 63 V, 100 V preferred. The shop lists no voltage rating, so CHECK THE PRINTING ON THE ACTUAL PARTS before fitting them.

3. Series pairs: $R101+R102 = 9.4 \text{ ohm} / 10 \text{ W}$ (design 10 ohm). $R201+R202 = 20 \text{ ohm} / 10 \text{ W}$ (20).

4. Parallel caps: $C102||C103 = 5.5\mu\text{F}$ (5.6). $C104||C105 = 10.1\mu\text{F}$ (10). $C201..C205 = 41\mu\text{F}$ (40).

$C206||C207 = 11.5\mu\text{F}$ (12). Each part's 'Design value' field holds the original.

5. L101/L102/L201/L202 are air-core coils, 0.8 mm (20 AWG) wire – self-wound or bought (Dayton

AC20-20 / AC20-25 / AC20-70 / AC202 or Jantzen Audio equivalents. Target DCR is in Description.

6. FOOTPRINTS ARE PROVISIONAL: measure cap lead pitch before routing. Coils mount OFF the board (lead pads only).

7. Single-sided board, CNC isolation-milled with a 0.8 mm end mill. Through-hole only.

Sheet: /

File: illuminate7mk2-crossover.kicad_sch

Title: illuminate 7 Mk2 – passive crossover (one per speaker)

Size: A4

Date:

Rev:

KiCad E.D.A. 10.0.6

Id: 1/1